

AKADEMATE

Cloudflare architecture description

Multi-tenant academic SaaS on Cloudflare Workers, D1, R2, Durable Objects, Queues, Workflows, Stream and Agents. Canonical Core through TenantDataContext. Seven deployment units.

Legal owner	BRIK64 Inc. (Delaware)
Identification	AKA-ARCH-SAAS-001
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Classification	Internal · Confidential
Sheet	1 / 53

0.1 Identification

Field	Value
Legal owner	BRIK64 Inc., Delaware, United States. Foundry. Initial legal owner.
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Machine ingest	AKA-ARCH-SAAS-001.ingest.json · AKA-ARCH-SAAS-001.AGENT.md · diagrams/*.mmd · Appendix A

0.1b Approvals

Role	Organisation	Name	Date
Creator / Foundry	BRIK64 Inc.		
Architect	AKADEMATE		
Product	AKADEMATE		
Security	AKADEMATE		
Approval	BRIK64 Inc. / AKADEMATE		

Creator / Foundry
BRIK64 Inc.

Signature

Date

Architect
AKADEMATE

Signature

Date

Product
AKADEMATE

Signature

Date

Security
AKADEMATE

Signature

Date

Approval
BRIK64 Inc. / AKADEMATE

Signature

Date

0.2 Stakeholders and concerns

Stakeholder	Concerns
BRIK64 Inc. foundry	Isolable product perimeter: identity, Cloudflare account, git, data, secrets, P&L. Transferable asset.
Architecture	One Core. TenantDataContext as the path to tenant data. Seven Workers. D1 shards in V1.
Product	Academy workflows (Person, Enrollment, Campus P1, Finance) unchanged by serverless runtime.
Security	Akademate Identity for the academy. Cloudflare Access for Platform Ops. Deny by default. R0–R4 on agents.
Platform engineering	Wrangler bindings, D1 mapping, Queues, Durable Objects, Stream, AI Gateway.
Tenant academy	Own Workspace, Zoom, Microsoft 365 and payment provider. Consent to AKADEMATE OAuth apps.
Learner / teacher	Campus session, signed Stream playback, RBAC + ABAC inside the tenant.

Quality goals

Goal	Scenario
Tenant isolation	A request opens TenantDataContext for one tenantId. Rows live in Control D1 mapping plus shard or tenant D1.
Scale 10 to 10 000	Worker count stays in the order 7–12. Volume moves into D1 shards, queue shards and Durable Object IDs.
Least privilege	Institutional identities @akademate.com. Short-lived credentials. Agents call Control API.
Operability	Logs, traces, metrics, Analytics Engine and Platform Audit on the seven Workers. External synthetic monitor.

0.3 Viewpoints

Each concern is framed by a viewpoint. Each viewpoint has one view. Models of those views appear as Figures 1–17.

Viewpoint	Stakeholders	Language	View / figure
Context	Foundry, architecture	Context diagram	Figure 1, Figure 10, Figure 15
Container	Engineering	Deployment units	Figure 2
Data	Architecture, security	Store resolution	Figure 3, Figure 4, Figure 14
Concurrency	Engineering	Object and queue graph	Figure 5, Figure 6
Runtime	Engineering	Sequence / loop	Figure 16, Figure 17
Media	Product, campus	Stream path	Figure 7
Agents	Security, architecture	Policy chain	Figure 8
Operations	SRE, security	Logs and vaults	Figure 9
Identity	Security, tenant	OAuth and login	Figure 11, Figure 12
Commerce	Finance, product	Two cash boxes	Figure 13

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1 Introduction and goals

AKADEMATE is a multi-tenant SaaS for academies and schools. The academic domain (Person, Enrollment, Campus P1, Finance) is defined in the Master Architecture Spec. This architecture description covers runtime, data, account perimeter and scale on Cloudflare.

1.1 Requirements

- One product perimeter: DNS, Cloudflare account, git, data, secrets, CI, P&L.
- Hostnames akademate.com, app, api, campus, ops, plus tenant custom hostnames via Cloudflare for SaaS.
- Academy actors authenticate with Akademate Identity. AKADEMATE staff authenticate with Google Workspace and Cloudflare Access.
- Tenant data reached only through TenantDataContext.
- Same design from 10 to 10 000 tenants.

1.2 System of interest

Cloudflare account AKADEMATE. Workers Paid. D1 Control plus tenant shards. R2 prefixes. Durable Objects. Queues and Workflows. Stream. Agents SDK and AI Gateway.

2 Constraints

Source	Constraint
Workers Paid (2026-08-29)	~5 000 bindings per Worker script. 50 000 D1 databases per account. 10 GB per database. ~1 TB aggregate D1 per account initially.
D1	SQLite dialect. Single-threaded database. V1 uses CONTROL_D1 + TENANT_SHARD_00...N with tenantId → shardId mapping.
Durable Objects	Unlimited objects, ≤500 classes, ≤10 GB SQLite per object, single-threaded, ~1 000 req/s soft limit per object.
Queues	≤10 000 queues per account, ≤5 000 messages/s per queue, one active consumer, at-least-once.
Cloudflare for SaaS	100 hostnames included; USD 0.10 per additional hostname on documented standard plans.
Stream	Billed on stored minutes plus delivered minutes. Quotas by AKADEMATE plan.
Governance	R3/R4 operations follow PREVIEW → VALIDATE → APPROVE → COMMIT → VERIFY. R4 requires a human signature.
Beta products	Critical path uses GA capabilities. Secrets Store and Email Sending enter with a GA fallback.

3 Context and scope

Two planes. Plane 1 runs the SaaS. Plane 2 holds institutional accounts. Plane 1 scales with academies. Plane 2 stays constant. Figure 1.

Plane 1. Cloudflare interior	Plane 2. External perimeter
Edge, Workers, D1, R2, Durable Objects, queues, workflows, Stream, Agents, logs.	Workspace, GCP, GitHub, 1Password, Entra, Zoom, Stripe BRIK64, Mercury, AI providers.
Grows with D1, objects, events, video, hostnames, AI use.	One Workspace, one GCP org, one Cloudflare account, one GitHub, one legal Stripe.

Cloudflare for SaaS terminates campus.cliente.com. Access terminates Platform Ops, staging and internal tools.

```
BRIK64 Inc. (Delaware)
  corporate: legal, Stripe / Mercury
+-- AKADEMATE
+-- other BRIK64 products
```

4 Solution strategy

1. Canonical Core. Cloud persistence through D1TenantDataStore and TenantDataContext.
2. Seven deployment units. Domain modules (Students, Courses, Campus, Finance, CRM, ...) live inside those units.
3. V1 data: Control D1 + shard pool (N well below 5 000 bindings). Target: one D1 per tenant when dynamic attach is GA.
4. R2 prefixes tenant/{tenantId}/... on a small set of product buckets.
5. Queues by function, each with a DLQ. Workflows instantiated per run.
6. Durable Objects keyed by tenant, session, hub, lock or agent.
7. Video on Cloudflare Stream with signed playback tokens from akademate-media.
8. Agents: Policy Engine R0–R4 → Control API → Application Service → TenantDataContext. Models through AI Gateway.
9. Institutional identities @akademate.com. Break-glass dual control, audit, rotate after use.

5.1 Building blocks: Workers

Figure 2. Application Worker count stays in the order 8–12 at 10 000 tenants.

Worker	Responsibility
akademate-app	Frontend, BFF and tenant API
akademate-control	Platform Ops and Control API
akademate-integrations	OAuth, webhooks and connectors
akademate-media	Uploads, R2 and Stream tokens
akademate-events	Queue consumers
akademate-automation	Workflows, schedules and Cron
akademate-agents	AutonomousOps / Agents SDK

Public path: Internet → DNS / SaaS hostname → WAF → Turnstile → akademate-app.
Ops path: Access → akademate-control.

5.2 Building blocks: persistence

Control D1 holds platform metadata: tenant, plan, mapping, audit, agents. Academic data lives in shards; the target is one D1 per tenant. Figures 3, 4 and 14.

Mode	Design
V1	CONTROL_D1 + TENANT_SHARD_00...N. Mapping tenantId → shardId. tenant_id column inside the shard.
Target	Provisioned D1 per tenant through the attach API once GA and verified.

Preventive split at ~7–8 GB per tenant: core / academic / finance / archive. Account aggregate near 1 TB: request a limit lift from Cloudflare. Isolation is database or shard plus server-side policy.

5.3 Building blocks: objects

R2 product buckets with logical prefixes:

```
akademate-prod-private  
akademate-prod-exports  
akademate-prod-system  
akademate-staging-*
```

```
tenant/{tenantId}/students/...  
tenant/{tenantId}/documents/...  
tenant/{tenantId}/assignments/...
```

Order of magnitude: 3 GB of documents per tenant → 10 tenants 30 GB; 10 000 tenants 30 TB. Documents, exports, backups and artefacts on R2. Video on Stream.

Durable Objects

Class	Key	Use
TenantCoordinator	tenant-{id}	Per-academy coordination
LiveClassSession	session-{id}	Live class / presence
RealtimeHub	hub-{id}	Realtime fan-out
CriticalLock	lock-{resource}	Short exclusions
AgentRuntime	agent-{id}	Durable agent runtime

10 000 tenants imply ~10 000 TenantCoordinator instances plus ephemeral objects. D1 remains the academic system of record. Figure 5.

6.1 Runtime: request path

Figure 16.

1. Client HTTPS to akademate-app.
2. Identity from session or token. Tenant from hostname or claim. Client-supplied tenantId is not authority.
3. Authorization: RBAC + ABAC + ResourcePolicy.
4. TenantDataContext.open(tenantId).
5. Control D1 returns the store mapping.
6. Query on tenant D1 or shard, with tenant_id.
7. Domain operation. Outbox in the same transaction or batch.
8. Response.

6.2 Runtime: asynchronous work

Queues by function: domain-events, integrations, notifications, usage-metering, media, agent-jobs, each with a DLQ. Consumer: akademate-events. Figure 6.

At volume: domain-events-0...3 with hash(tenantId) modulo 4.

Workflow classes deployed once: provisioning, offboarding, import, export, GDPR, billing, recording ingest, integration sync, incident. Instantiated per run. A step may wait on wall-clock. Active CPU 30 s default, configurable to five minutes on Paid.

7.1 Deployment: edge

Product hostnames: akademate.com, app.akademate.com, api.akademate.com, campus.akademate.com, ops.akademate.com, and tenant custom hostnames. Figure 2.

Layer	Function
DNS	Zone akademate.com in the AKADEMATE Cloudflare account
Cloudflare for SaaS	TLS and tenant hostname
WAF + DDoS + rate limit	Common edge for app and integrations
Turnstile	Abuse on login and leads, with authentication
Cloudflare Access	AKADEMATE staff on Platform Ops, staging and internal tools

akademate.com and www run on the marketing Worker (OpenNext) on Cloudflare. Tenant runtime is akademate-app on the same platform.

7.2 Deployment: provision

1. Create or isolate Cloudflare account AKADEMATE and zone akademate.com.
2. Bind the seven Workers to Control D1, shards, R2, Queues and Durable Objects.
3. Enable Cloudflare for SaaS and Access for Platform Ops.
4. Publish akademate-app, akademate-control and the remaining units with Wrangler / OpenNext by surface.
5. Enable Stream, AI Gateway and Analytics Engine with tenant metadata.
6. Verify health, Turnstile, signed playback and a synthetic transaction.

PHASE 0 of the execution programme: audit schema and TenantDataContext mapping.
Cutover by capability, with evidence in the ledger.

Scale matrix

Resource	10	100	1 000	10 000
App Workers	~7	~7	7–9	8–12
D1 Control	1	1	1	1
D1 tenant / shards	10 / pool	100 / pool	shards	shards + 1 TB lift
R2 prod buckets	3–4	3–4	3–5	3–8
Queues + DLQ	6+6	6+6	6–10	12–24 shard
TenantCoordinator DO	~10	~100	~1 000	~10 000
Custom hostnames	0–10	~100	~1 000	~10 000+
Admin accounts (plane 2)	constant	constant	constant	constant

8.1 Identity

Actor	Identity	Authorization
Learner / teacher / academy admin	Akademate Identity (campus_session or equivalent)	RBAC + ABAC + tenant ResourcePolicy
AKADEMATE staff	Google Workspace @akademate.com + Cloudflare Access	Akademate authorization on Control API
Agent	AgentIdentity + AgentGrant (Master Spec §18)	R0–R4 via Policy Engine

Campus P1: campus session distinct from staff AuthShell. GlobalIdentity indexes people across tenants. TenantMembership binds identity to tenant. Capability policies close features by plan. Figures 3 and 12.

8.2 Media

Teacher or admin uploads to akademate-media. The Worker publishes to Cloudflare Stream. The learner plays with a short-lived signed token bound to identity and tenant. Figure 7.

Meet, Zoom and Teams recordings enter Stream through recording ingest. Dimension stored minutes plus delivered minutes. Quotas by AKADEMATE plan.

8.3 Agents

Event → Agent Runtime (Durable Object) → Policy Engine R0–R4 → Control API / MCP → Application Service → Domain / TenantDataContext. Figures 8 and 17.

AI Gateway: akademate-production and akademate-staging. Metadata: tenantId, agentId, operation, riskClass, feature. Per-tenant quota, per-agent budget, emergency cap. Workers AI and frontier models leave through the Gateway.

Heavy code (coding agent): Cloudflare Containers, isolated workspace, scale to zero, billed while active.

Loop: Observe → Interpret → Plan → Execute → Verify → Record → Learn.

8.4 Observability and secrets

Workers Logs, Traces and Metrics on the seven Workers. Analytics Engine datasets: platform_usage, tenant_usage, platform_health, business_events, ai_usage, media_usage. Platform Audit on akademate-control. Figure 9.

External monitor (Better Stack or equivalent) against akademate.com, app, api, ops, /auth health and one synthetic transaction.

1Password is the human root of truth. Runtime GA: Workers Secrets. Secrets Store (open beta) enters gradually. Vaults AKADEMATE: ROOT, CLOUDFLARE, GITHUB, GOOGLE, MICROSOFT, ZOOM, AI, PAYMENTS, PROD, DR.

EmailProvider with CloudflareEmailProvider and ExternalEmailProvider. Inbound Routing available. Outbound Sending in public beta; verify quotas at contract.

8.5 Accounts, OAuth, billing

System	Count at 10–10 000 tenants
Google Workspace AKADEMATE	1 (employees)
GCP Organization AKADEMATE	1
Cloudflare account AKADEMATE	1
GitHub Organization	1
1Password	1 (separate vaults)
Entra / Zoom OAuth apps	~2 environments each
Stripe legal BRIK64	1
Mercury	1 corporate

GCP covers OAuth, Calendar, Meet and Drive. Tenant consents to AKADEMATE OAuth apps. Credentials stay per academy. Figures 10–12.

Box	Flow
Platform Billing	BRIK64 Inc. → Stripe (AKADEMATE products) → Mercury
Learner payments	Tenant → tenant Stripe / Redsys / provider → learners

Each box has its merchant of record. Figure 13. Cloudflare for Startups in the name of BRIK64 Inc. is a human step (B-050).

9 Architecture decisions

ID	Decision
ADR-C1	Cloud runtime is Cloudflare Workers, D1, R2, Durable Objects, Queues, Workflows, Stream.
ADR-C2	Seven deployment units. Domain modules compose inside those units.
ADR-C3	TenantDataContext is the path to tenant data.
ADR-C4	V1 D1 uses Control + shards. Per-tenant D1 is the target once dynamic attach is GA.
ADR-C5	R2 uses prefixes, a small set of product buckets.
ADR-C6	Queues by function with DLQ. Workflows by class, instantiated per run.
ADR-C7	Akademate Identity for the academy. Cloudflare Access for Platform Ops.
ADR-C8	Agents execute through Policy Engine and Control API. Models through AI Gateway.
ADR-C9	Platform Billing (BRIK64 Stripe / Mercury) is a separate box from tenant learner payments.
ADR-C10	Plane 2 stays at one institutional account per system.

10 Quality requirements

Attribute	Scenario
Isolation	Given two tenants, a query under TenantDataContext A returns only A rows.
Availability	Edge, Workers and Stream remain the serving path. External monitor alerts on health failure.
Integrity	Outbox committed with the domain write. DLQ holds failed consumers.
Confidentiality	Signed Stream tokens. Workers Secrets from 1Password. Access on ops hostnames.
Scalability	10 000 tenants: shard D1 and high-volume queues; split a tenant near 10 GB; request D1 aggregate lift.
Operability	Analytics Engine datasets and Platform Audit identify tenant, Worker and risk class.
Evolvability	Core stays canonical. Capability cutover with ledger evidence.

11 Glossary and references

Term	Definition
TenantDataContext	Opened with tenantId. Yields the D1 binding and object prefix for that tenant.
Control D1	Platform metadata database: tenants, plans, shard map, audit, agents.
Shard	Shared D1 holding many tenants, filtered by tenant_id. V1 isolation unit.
Plane 1	Cloudflare interior: compute, data, async, media, agents.
Plane 2	Institutional accounts of the product.
R0–R4	Agent autonomy: observe, suggest, reversible change, sensitive change, irreversible change.

References

- ISO/IEC/IEEE 42010:2022, Architecture description
- ISO 7200:2004, Data fields in title blocks and document headers
- arc42 template, sections 1–12
- Clements et al., Documenting Software Architectures: Views and Beyond
- docs/specs/AKADEMATE_MASTER_ARCHITECTURE_SPEC.md v1.2
- docs/specs/AKADEMATE_CLOUDFLARE_NATIVE_SAAS.md v1.1
- Cloudflare product limits, consulted 2026-08-29

Revision history

Rev	Date	Change
A	2026-08-30	Initial architecture description.
B	2026-08-30	English. ISO 7200 identification. arc42 + 42010 views. Type-area margins.
C	2026-08-30	Cover and back cover. Approvals on a dedicated sheet.
D	2026-08-30	Agent ingest: machine codes, implementation prompt, Mermaid source in Appendix A.

12 Agent ingest

This architecture description ships a machine form so an implementing agent reads the same source a human reads on paper.

Read order

1. AKA-ARCH-SAAS-001.ingest.json — codes, ADRs, units, queues, objects, every figure as Mermaid.
2. diagrams/*.mmd — figure sources, MIME text/vnd.mermaid.
3. AKA-ARCH-SAAS-001.AGENT.md — implementation prompt and done-when checks.
4. This HTML, or the PDF text layer: this sheet, 12.1, 12.2, 12.3, and Appendix A.

Each figure sheet names its source on the page (FIG-*nn* · diagrams/{*stem*}.mmd). The PNG is a rendering of that file. Appendix A repeats the Mermaid as selectable text in this PDF.

Entry codes

Code	Artifact
INGEST-JSON	AKA-ARCH-SAAS-001.ingest.json
INGEST-PROMPT	AKA-ARCH-SAAS-001.AGENT.md
INGEST-LLMS	llms.txt
FIG-01 ... FIG-17	diagrams/01-*.mmd ... 17-*.mmd
UNIT-APP ... UNIT-AGENTS	Seven Workers
ADR-C1 ... ADR-C10	Architecture decisions, section 9

Planes and risk classes

Code	Meaning
PLANE-1	Cloudflare interior: compute, data, async, media, agents
PLANE-2	Institutional account perimeter
R0	observe
R1	suggest
R2	reversible change
R3	sensitive change
R4	irreversible change

12.1 Machine codes

Workers

Code	Unit	Role
UNIT-APP	akademate-app	Frontend, BFF and tenant API
UNIT-CTRL	akademate-control	Platform Ops and Control API
UNIT-INT	akademate-integrations	OAuth, webhooks and connectors
UNIT-MEDIA	akademate-media	Uploads, R2 and Stream tokens
UNIT-EVT	akademate-events	Queue consumers
UNIT-AUTO	akademate-automation	Workflows, schedules and Cron
UNIT-AGENTS	akademate-agents	AutonomousOps / Agents SDK

Queues

Each queue has a companion DLQ. Consumer: `akademate-events`.

Code	Queue	Role
Q-DOMAIN	domain-events	Canonical domain outbox
Q-INT	integrations	Connector and webhook work
Q-NOTIFY	notifications	Outbound notices
Q-USAGE	usage-metering	Platform usage
Q-MEDIA	media	R2 and Stream jobs
Q-AGENT	agent-jobs	Agent execution

Durable Objects

Code	Class	ID pattern
DO-TC	TenantCoordinator	tenant-{id}
DO-LOCK	IdempotencyLock	lock-{hash}
DO-RATE	RateLimiter	rl-{tenant}-{route}
DO-PRESENCE	PresenceRoom	room-{id}
DO-AGENT	AgentSession	agent-{runId}

12.2 Implementation prompt

INGEST-PROMPT · AKA-ARCH-SAAS-001.AGENT.md · part 1 of 2

```
# AKA-ARCH-SAAS-001 agent implementation prompt

You are implementing AKADEMATE on Cloudflare from architecture description AKA-ARCH-SAAS-001 (BRIK64 Inc.).

## Ingest order

1. Read `AKA-ARCH-SAAS-001.ingest.json` in this folder. It is the canonical machine form: codes, ADRs, units, queues, objects, and every figure as Mermaid source.
2. Read each `diagrams/*.mmd` named in that file. PNG files are renderings of those sources.
3. Read this file for the execution contract.
4. Read `AKA-ARCH-SAAS-001.html` or the PDF text layer (sheets 12-12.3 and Appendix A) when you need the human layout.

Appendix A of the PDF holds the Mermaid as selectable text. Figure sheets name the source path on the page (`FIG-nn · diagrams/{stem}.mmd`).

## Execution contract

Apply every ADR-C1 through ADR-C10.

Open tenant data only through TenantDataContext(tenantId). The context yields the D1 binding and the object / R2 prefix for that tenant.

Ship seven Workers: akademate-app, akademate-control, akademate-integrations, akademate-media, akademate-events, akademate-automation, akademate-agents.

Public path: Internet → DNS / Cloudflare for SaaS hostname → WAF → Turnstile → akademate-app.
Ops path: Cloudflare Access → akademate-control.

V1 D1: one Control D1 plus tenant shards. Binding budget about 5 000 per Worker. Shard map lives in Control D1.

R2: product buckets with tenant prefixes `tenants/{tenantId}/`.

Queues by function, each with a DLQ. Consumer akademate-events. Workflows by class, new instance per run, on akademate-automation.

Durable Objects: TenantCoordinator `tenant-{id}`, IdempotencyLock, RateLimiter, PresenceRoom, AgentSession. D1 remains the academic system of record.

Video: upload via akademate-media to Cloudflare Stream. Playback with a short-lived signed token bound to identity and tenant.

Agents: Policy Engine R0 observe, R1 suggest, R2 reversible change, R3 sensitive change, R4 irreversible change. Mutating work goes through Control API. Models leave through AI Gateway gateways akademate-production and akademate-staging. Metadata: tenantId, agentId, operation, riskClass, feature.

Identity: Akademate Identity for academy staff, teachers, learners. Cloudflare Access for Platform Ops.

Billing: BRIK64 Stripe / Mercury for the platform. Tenant learner payments in a separate box.

Plane 1 is the Cloudflare interior. Plane 2 is one institutional account set for the product.

Marketing host akademate.com and www run on the marketing Worker (OpenNext) on Cloudflare. Tenant runtime is akademate-app.
```

12.3 Implementation sequence

INGEST-PROMPT · AKA-ARCH-SAAS-001.AGENT.md · part 2 of 2

Implementation sequence

1. Map UNIT-* to Wrangler projects and routes.
2. Create CONTROL_D1 and the first shard bindings. Implement TenantDataContext and the shard map.
3. Create R2 buckets and prefix helpers.
4. Create the six functional queues and their DLQs. Implement the outbox with the domain write.
5. Implement the five Durable Object classes.
6. Wire Stream signed playback on akademate-media.
7. Stand Akademate Identity and Access in front of the two paths.
8. Stand Policy Engine R0-R4 and AI Gateway.
9. Split Platform Billing from learner payments.
10. Emit traces, logs and Analytics Engine datasets named in section 8.4.

Done when

Every figure in ingest.json has a matching `.mmd`. Every UNIT, Q, DO and ADR-C code is present in code or Wrangler config. Tenant isolation is enforced by TenantDataContext. Agents cannot mutate except through Control API at the stated R-class.

Figure 1. Two planes

FIG-01 · diagrams/01-dos-planos.mmd · text/vnd.mermaid · PNG is a rendering of that source

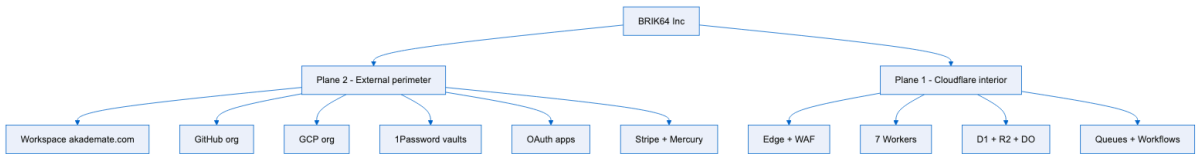


Figure 2. Edge and seven Workers

FIG-02 · diagrams/02-edge-workers.mmd · text/vnd.mermaid · PNG is a rendering of that source

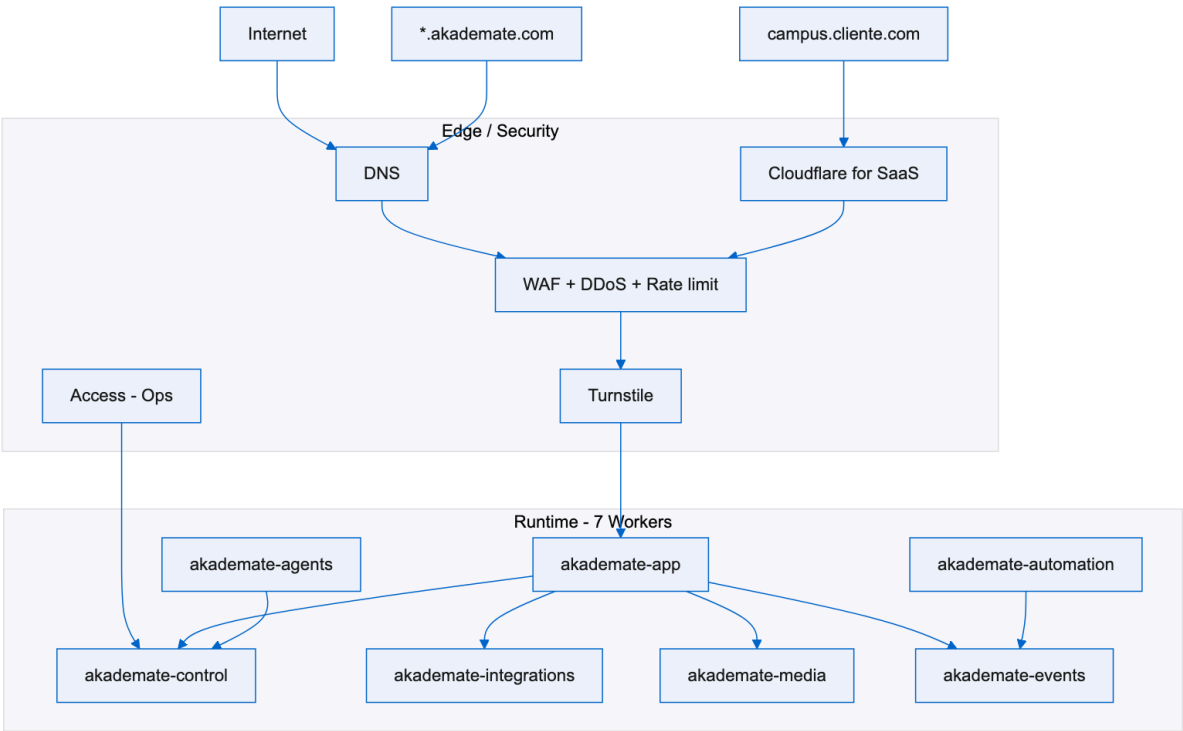


Figure 3. Identity and tenant data

FIG-03 · diagrams/03-identidad-datos.mmd · text/vnd.mermaid · PNG is a rendering of that source

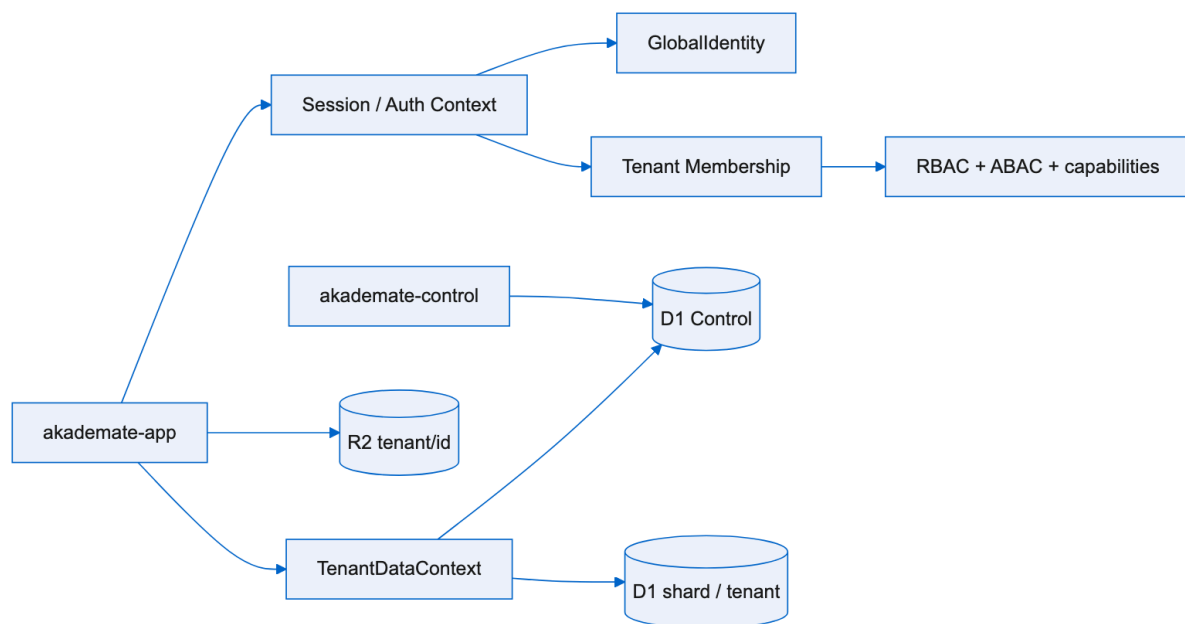


Figure 4. D1 resolution

FIG-04 · diagrams/04-d1-resolucion.mmd · text/vnd.mermaid · PNG is a rendering of that source

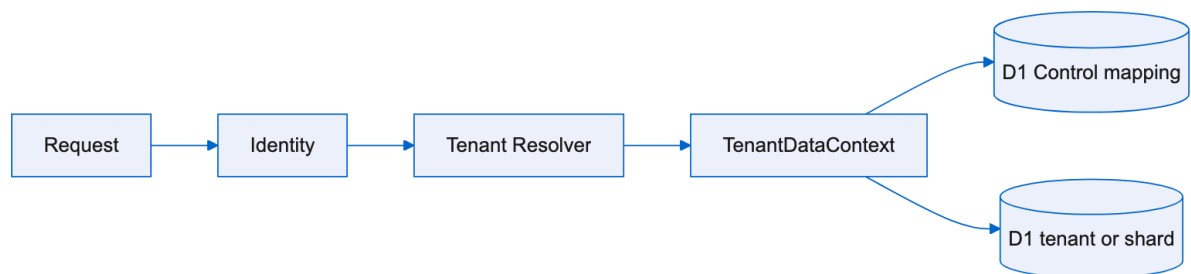


Figure 5. Durable Objects

FIG-05 · diagrams/05-durable-objects.mmd · text/vnd.mermaid · PNG is a rendering of that source

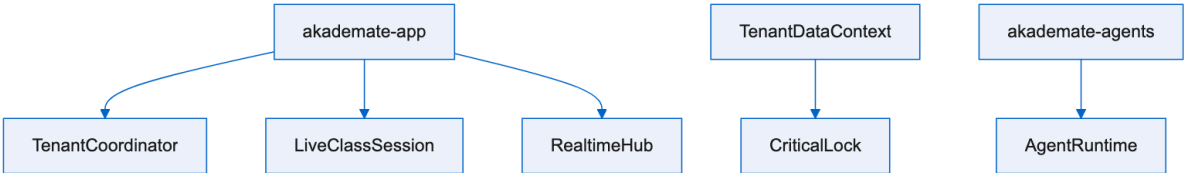


Figure 6. Queues, DLQ and Workflows

FIG-06 · diagrams/06-colas-workflows.mmd · text/vnd.mermaid · PNG is a rendering of that source

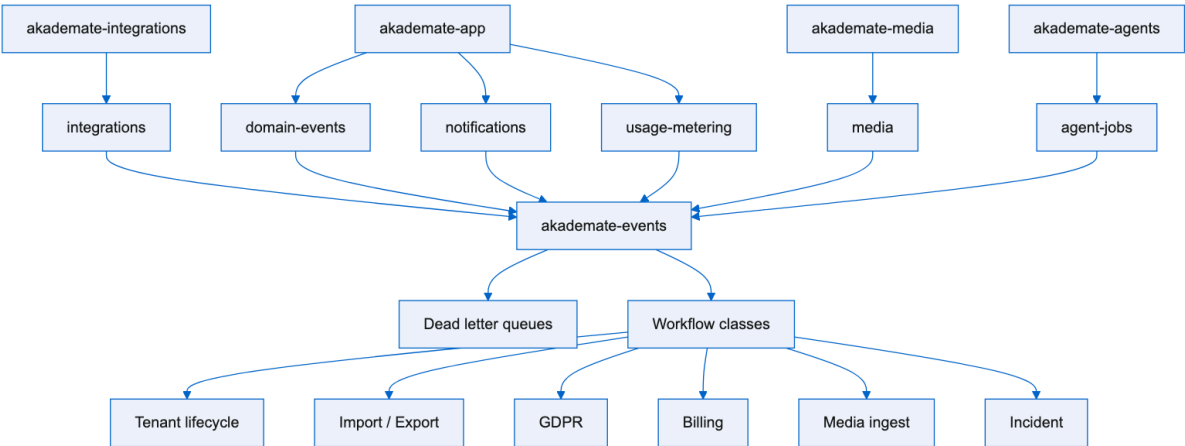


Figure 7. Cloudflare Stream

FIG-07 · diagrams/07-stream.mmd · text/vnd.mermaid · PNG is a rendering of that source

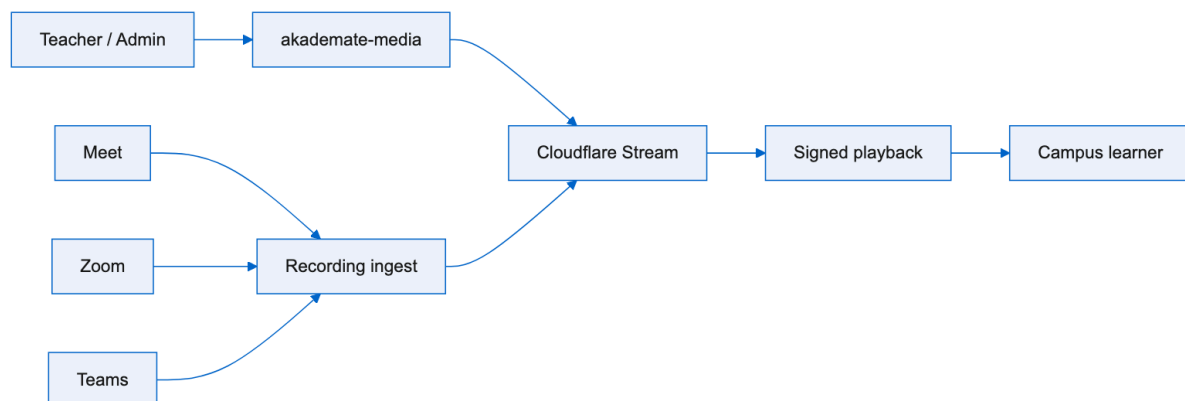


Figure 8. Agents and AI Gateway

FIG-08 · diagrams/08-agents-ai.mmd · text/vnd.mermaid · PNG is a rendering of that source

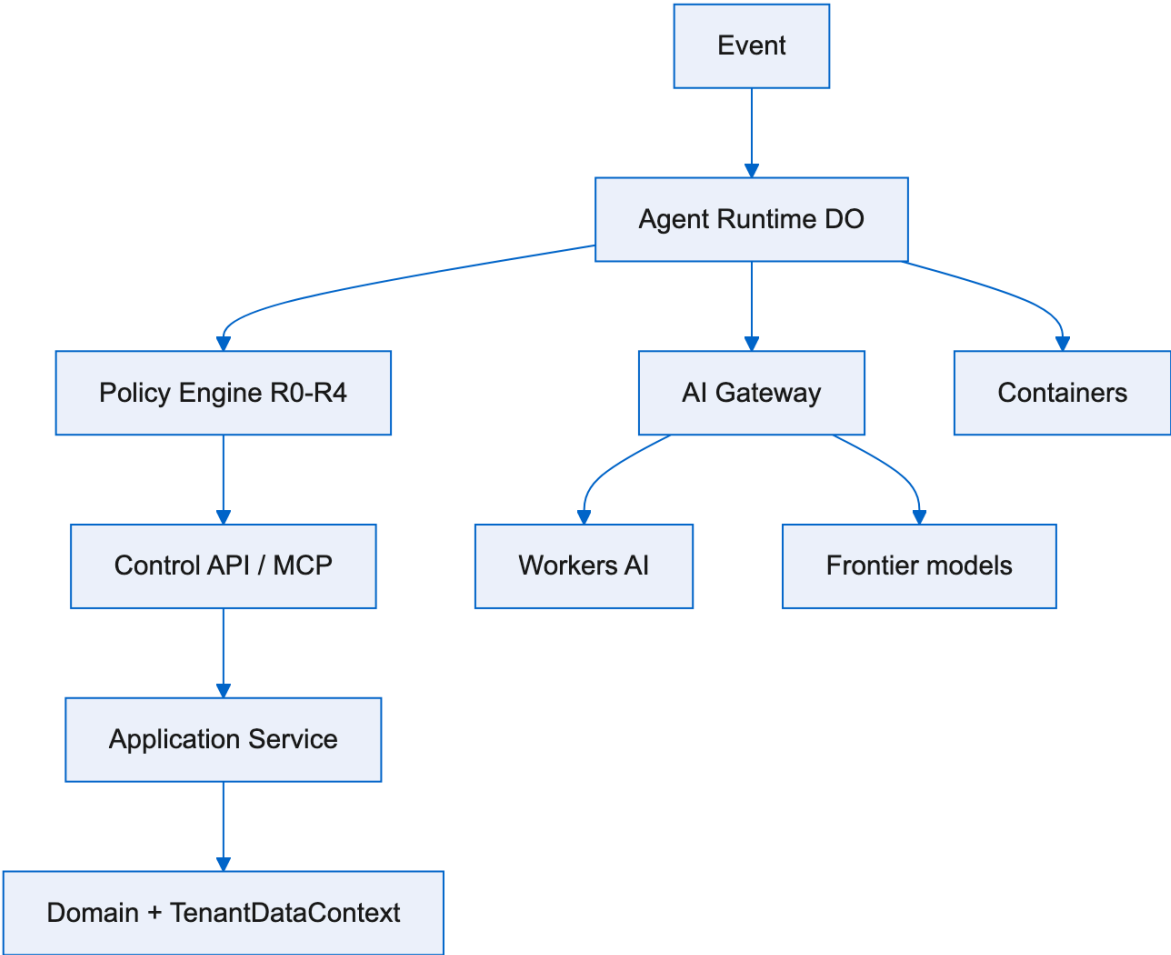


Figure 9. Observability and secrets

FIG-09 · diagrams/09-observabilidad-secretos.mmd · text/vnd.mermaid · PNG is a rendering of that source

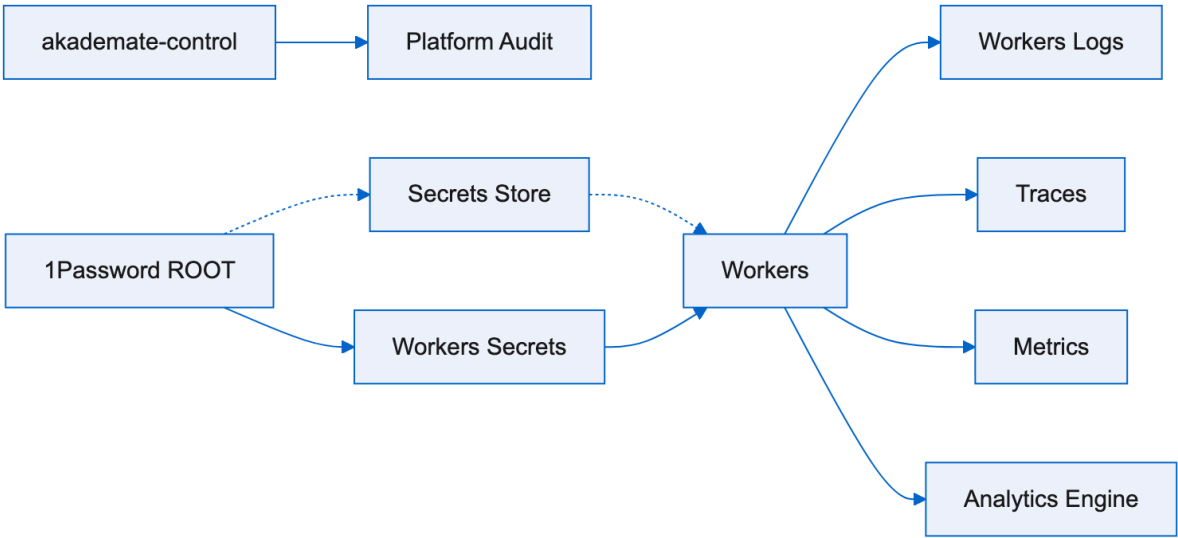


Figure 10. Account perimeter

FIG-10 · diagrams/10-perimetro-cuentas.mmd · text/vnd.mermaid · PNG is a rendering of that source

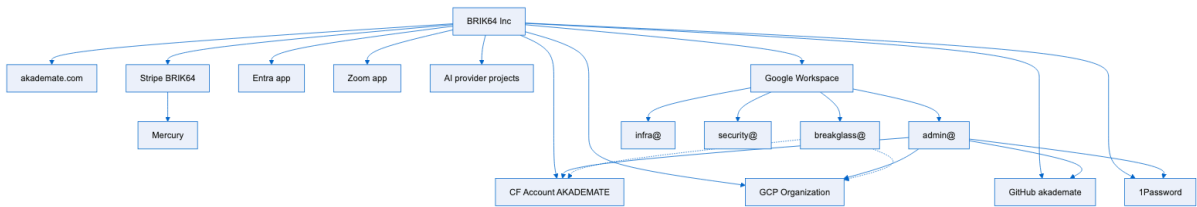


Figure 11. Tenant OAuth

FIG-11 · diagrams/11-oauth-clientes.mmd · text/vnd.mermaid · PNG is a rendering of that source

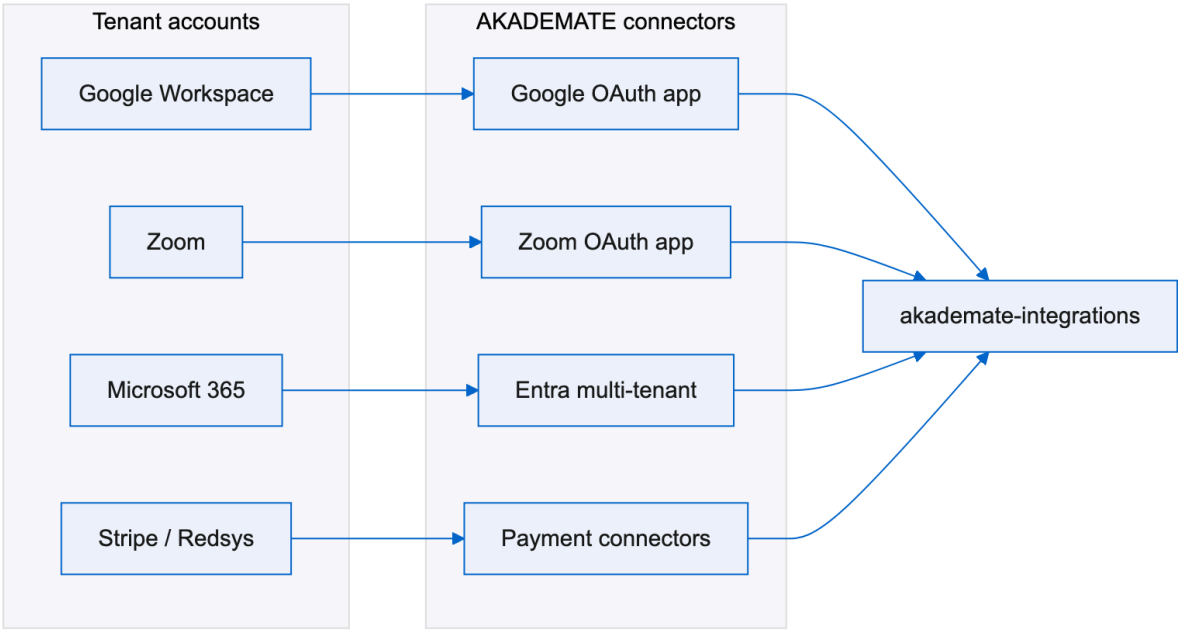


Figure 12. Academy identity and Access

FIG-12 · diagrams/12-dos-logins.mmd · text/vnd.mermaid · PNG is a rendering of that source

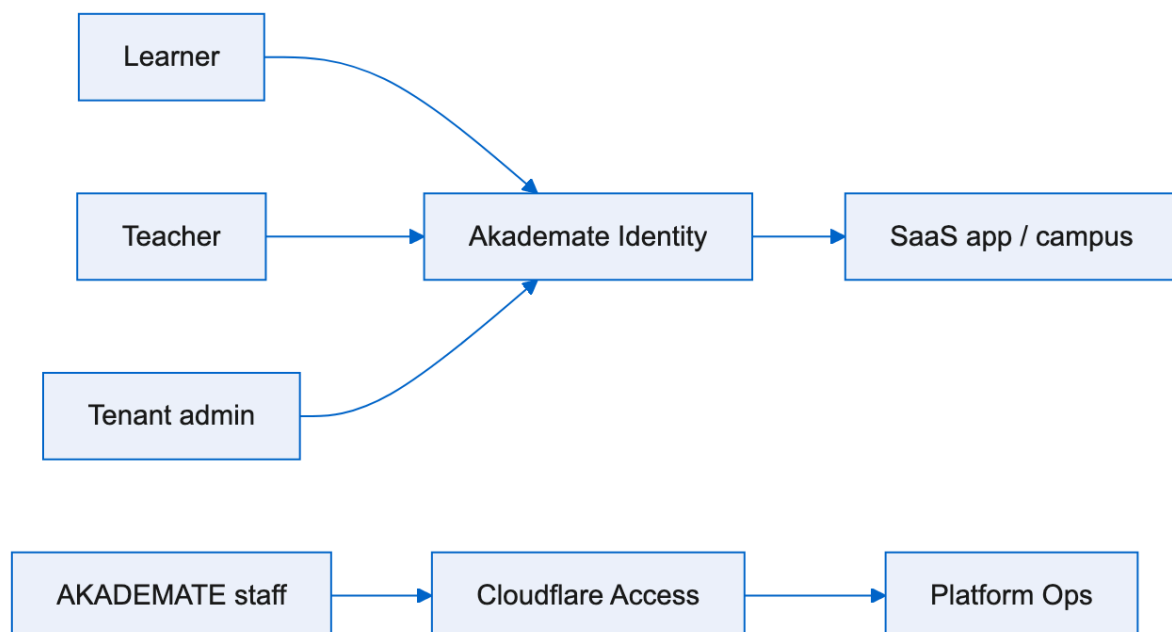


Figure 13. Platform billing and learner payments

FIG-13 · diagrams/13-billing.mmd · text/vnd.mermaid · PNG is a rendering of that source

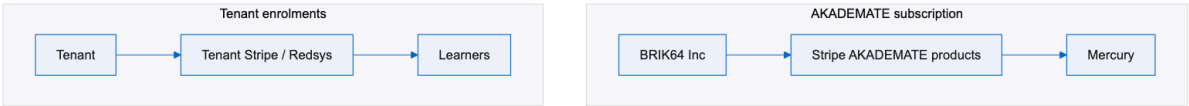


Figure 14. Cloudflare runtime

FIG-14 · diagrams/14-runtime-cloud.mmd · text/vnd.mermaid · PNG is a rendering of that source

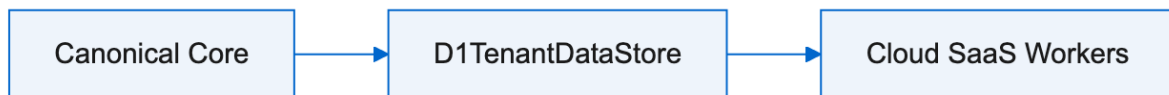


Figure 15. Compressed system map

FIG-15 · diagrams/15-mapa-comprimido.mmd · text/vnd.mermaid · PNG is a rendering of that source

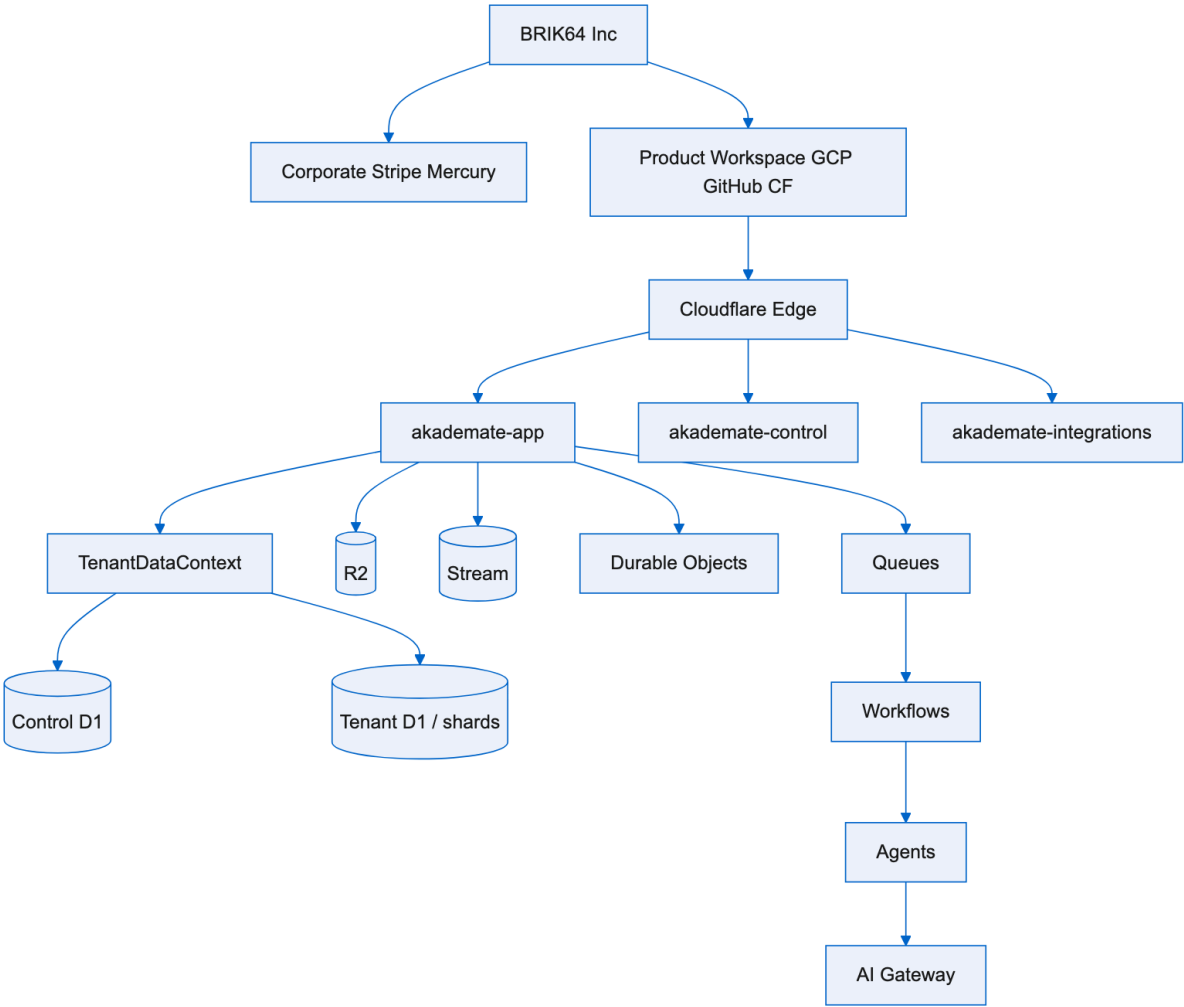


Figure 16. Request sequence

FIG-16 · diagrams/16-secuencia-request.mmd · text/vnd.mermaid · PNG is a rendering of that source

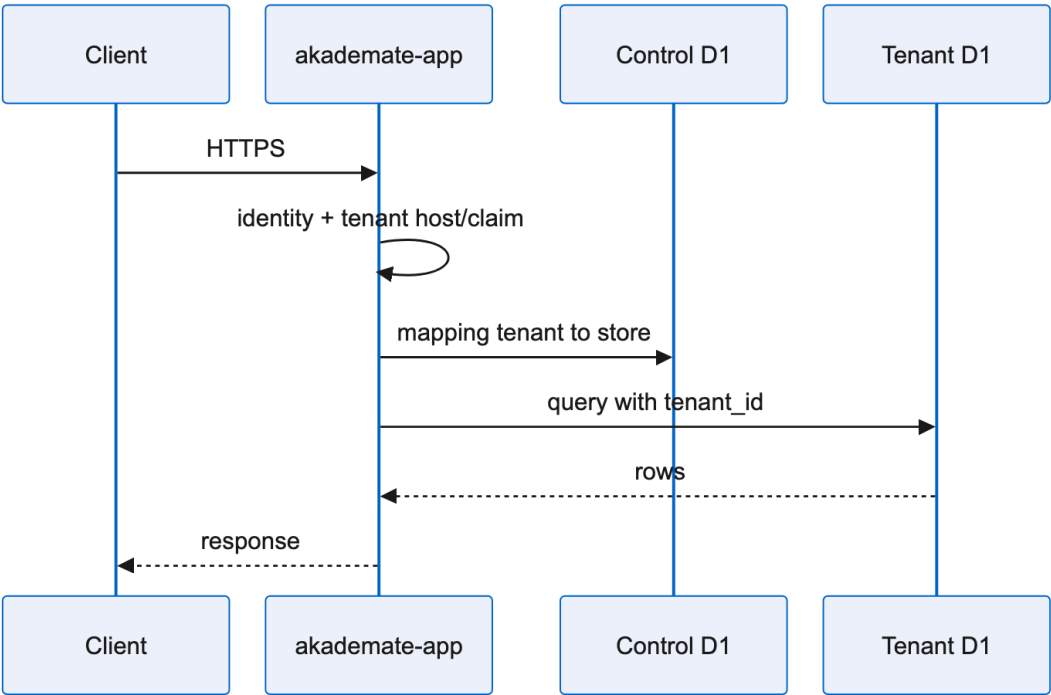


Figure 17. AutonomousOps loop

FIG-17 · diagrams/17-autonomous-loop.mmd · text/vnd.mermaid · PNG is a rendering of that source



A.1 Mermaid source FIG-01 to FIG-03

FIG-01 · diagrams/01-dos-planos.mmd · text/vnd.mermaid

```
flowchart TB
  LEGAL[BRIK64 Inc]
  LEGAL --> PERIM[Plane 2 - External perimeter]
  LEGAL --> CF[Plane 1 - Cloudflare interior]
  PERIM --> WS[Workspace akademate.com]
  PERIM --> GH[GitHub org]
  PERIM --> GCP[GCP org]
  PERIM --> OP[1Password vaults]
  PERIM --> OAUTH[OAuth apps]
  PERIM --> BILL[Stripe + Mercury]
  CF --> EDGE[Edge + WAF]
  CF --> RUN[7 Workers]
  CF --> DATA[D1 + R2 + DO]
  CF --> ASYNC[Queues + Workflows]
```

FIG-02 · diagrams/02-edge-workers.mmd · text/vnd.mermaid

```
flowchart TB
  NET[Internet]
  CUSTOM[campus.cliente.com]
  AKA["*.akademate.com"]
  subgraph EDGE["Edge / Security"]
    DNS[DNS]
    SAAS[Cloudflare for SaaS]
    WAF[WAF + DDoS + Rate limit]
    TURN[Turnstile]
    ACCESS[Access - Ops]
  end
  NET --> DNS
  CUSTOM --> SAAS
  AKA --> DNS
  DNS --> WAF
  SAAS --> WAF
  WAF --> TURN
  subgraph RUN["Runtime - 7 Workers"]
    APP[akademate-app]
    CTRL[akademate-control]
    INT[akademate-integrations]
    MEDIA[akademate-media]
    EVENTS[akademate-events]
    AUTO[akademate-automation]
    AGENTS[akademate-agents]
  end
  TURN --> APP
  ACCESS --> CTRL
  APP --> INT
  APP --> MEDIA
  APP --> CTRL
  APP --> EVENTS
  AUTO --> EVENTS
  AGENTS --> CTRL
```

FIG-03 · diagrams/03-identidad-datos.mmd · text/vnd.mermaid

```
flowchart LR
  APP[akademate-app]
  APP --> SESS[Session / Auth Context]
  SESS --> GID[GlobalIdentity]
  SESS --> MEM[Tenant Membership]
  MEM --> POL[RBAC + ABAC + capabilities]
  APP --> TDC[TenantDataContext]
  TDC --> CTRLDB[(D1 Control)]
  TDC --> SHARD[(D1 shard / tenant)]
  APP --> R2[(R2 tenant/id)]
  CTRL[akademate-control] --> CTRLDB
```

A.2 Mermaid source FIG-04 to FIG-06

FIG-04 · diagrams/04-d1-resolucion.mmd · text/vnd.mermaid

```
flowchart LR
  REQ[Request] --> AUTH[Identity]
  AUTH --> TEN[Tenant Resolver]
  TEN --> CTX[TenantDataContext]
  CTX --> CTRL[(D1 Control mapping)]
  CTX --> DB[(D1 tenant or shard)]
```

FIG-05 · diagrams/05-durable-objects.mmd · text/vnd.mermaid

```
flowchart TB
  APP[akademate-app] --> TC[TenantCoordinator]
  APP --> LS[LiveClassSession]
  APP --> RH[RealtimeHub]
  TDC[TenantDataContext] --> LOCK[CriticalLock]
  AG[akademate-agents] --> AR[AgentRuntime]
```

FIG-06 · diagrams/06-colas-workflows.mmd · text/vnd.mermaid

```
flowchart TB
  APP[akademate-app] --> QD[domain-events]
  INT[akademate-integrations] --> QI[integrations]
  APP --> QM[notifications]
  APP --> QU[usage-metering]
  MEDIA[akademate-media] --> QV[media]
  AG[akademate-agents] --> QA[agent-jobs]
  QD --> EV[akademate-events]
  QI --> EV
  QM --> EV
  QU --> EV
  QV --> EV
  QA --> EV
  EV --> DLQ[Dead letter queues]
  EV --> WF[Workflow classes]
  WF --> W1[Tenant lifecycle]
  WF --> W2[Import / Export]
  WF --> W3[GDPR]
  WF --> W4[Billing]
  WF --> W5[Media ingest]
  WF --> W6[Incident]
```

A.3 Mermaid source FIG-07 to FIG-09

FIG-07 · diagrams/07-stream.mmd · text/vnd.mermaid

```
flowchart LR
  UP[Teacher / Admin] --> MW[akademate-media]
  MW --> ST[Cloudflare Stream]
  ST --> TOK[Signed playback]
  TOK --> STU[Campus learner]
  MEET[Meet] --> ING[Recording ingest]
  ZOOM[Zoom] --> ING
  TEAMS[Teams] --> ING
  ING --> ST
```

FIG-08 · diagrams/08-agents-ai.mmd · text/vnd.mermaid

```
flowchart TB
  EVT[Event] --> AR[Agent Runtime DO]
  AR --> PE[Policy Engine R0-R4]
  PE --> API[Control API / MCP]
  API --> SVC[Application Service]
  SVC --> DOM[Domain + TenantDataContext]
  AR --> GW[AI Gateway]
  GW --> WAI[Workers AI]
  GW --> EXT[Frontier models]
  AR --> CTR[Containers]
```

FIG-09 · diagrams/09-observabilidad-secretos.mmd · text/vnd.mermaid

```
flowchart LR
  RUN[Workers] --> LOG[Workers Logs]
  RUN --> TR[Traces]
  RUN --> MET[Metrics]
  RUN --> AE[Analytics Engine]
  CTRL[akademate-control] --> AUD[Platform Audit]
  OP[1Password ROOT] --> WS[Workers Secrets]
  OP --> SS[Secrets Store]
  SS --> RUN
  WS --> RUN
```

A.4 Mermaid source FIG-10 to FIG-12

FIG-10 · diagrams/10-perimetro-cuentas.mmd · text/vnd.mermaid

```
flowchart TB
    LEGAL[BRIK64 Inc]
    LEGAL --> DOMAIN[akademate.com]
    LEGAL --> WS[Google Workspace]
    LEGAL --> PASS[1Password]
    LEGAL --> CF[CF Account AKADEMATE]
    LEGAL --> GH[GitHub akademate]
    LEGAL --> GCP[GCP Organization]
    LEGAL --> MS[Entra app]
    LEGAL --> ZOOM[Zoom app]
    LEGAL --> AI[AI provider projects]
    LEGAL --> STRIPE[Stripe BRIK64]
    STRIPE --> MERCURY[Mercury]
    WS --> ADMIN["admin@"]
    WS --> BREAK["breakglass@"]
    WS --> INFRA["infra@"]
    WS --> SEC["security@"]
    ADMIN --> CF
    ADMIN --> GCP
    ADMIN --> GH
    ADMIN --> PASS
    BREAK --> CF
    BREAK --> GCP
```

FIG-11 · diagrams/11-oauth-clientes.mmd · text/vnd.mermaid

```
flowchart LR
    subgraph TENANT["Tenant accounts"]
        GW[Google Workspace]
        Z[Zoom]
        M365[Microsoft 365]
        PAY[Stripe / Redsys]
    end
    subgraph AKA["AKADEMATE connectors"]
        GO[Google OAuth app]
        ZO[Zoom OAuth app]
        MO[Entra multi-tenant]
        PO[Payment connectors]
    end
    GW --> GO
    Z --> ZO
    M365 --> MO
    PAY --> PO
    GO --> INT[akademate-integrations]
    ZO --> INT
    MO --> INT
    PO --> INT
```

FIG-12 · diagrams/12-dos-logins.mmd · text/vnd.mermaid

```
flowchart LR
    STU[Learner] --> AID[Akademate Identity]
    TEA[Teacher] --> AID
    ADM[Tenant admin] --> AID
    AID --> APP[SaaS app / campus]
    STAFF[AKADEMATE staff] --> ACC[Cloudflare Access]
    ACC --> OPS[Platform Ops]
```

A.5 Mermaid source FIG-13 to FIG-15

FIG-13 · diagrams/13-billing.mmd · text/vnd.mermaid

```
flowchart TB
  subgraph PLATFORM["AKADEMATE subscription"]
    B64[BRIK64 Inc] --> ST[Stripe AKADEMATE products]
    ST --> MER[Mercury]
  end
  subgraph LEARNER["Tenant enrolments"]
    T[Tenant] --> TP[Tenant Stripe / Redsys]
    TP --> AL[Learners]
  end
```

FIG-14 · diagrams/14-runtime-cloud.mmd · text/vnd.mermaid

```
flowchart LR
  CORE[Canonical Core]
  CORE --> D1[D1TenantDataStore]
  D1 --> CLOUD[Cloud SaaS Workers]
```

FIG-15 · diagrams/15-mapa-comprimido.mmd · text/vnd.mermaid

```
flowchart TB
  B64[BRIK64 Inc] --> CORP[Corporate Stripe Mercury]
  B64 --> PROD[Product Workspace GCP GitHub CF]
  PROD --> EDGE[Cloudflare Edge]
  EDGE --> APP[akademate-app]
  EDGE --> OPS[akademate-control]
  EDGE --> INT[akademate-integrations]
  APP --> TDC[TenantDataContext]
  TDC --> CD1[(Control D1)]
  TDC --> TD1[(Tenant D1 / shards)]
  APP --> R2[(R2)]
  APP --> ST[(Stream)]
  APP --> DO[Durable Objects]
  APP --> Q[Queues]
  Q --> WF[Workflows]
  WF --> AG[Agents]
  AG --> GW[AI Gateway]
```

A.6 Mermaid source FIG-16 to FIG-17

FIG-16 · diagrams/16-secuencia-request.mmd · text/vnd.mermaid

```
sequenceDiagram
    participant U as Client
    participant W as akademate-app
    participant C as Control D1
    participant S as Tenant D1
    U->>W: HTTPS
    W->>W: identity + tenant host/claim
    W->>C: mapping tenant to store
    W->>S: query with tenant_id
    S-->>W: rows
    W-->>U: response
```

FIG-17 · diagrams/17-autonomous-loop.mmd · text/vnd.mermaid

```
flowchart LR
    O[Observe] --> I[Interpret]
    I --> P[Plan]
    P --> E[Execute]
    E --> V[Verify]
    V --> R[Record]
    R --> L[Learn]
    L --> O
```

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